UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

General Certificate of Education O Level

MARK SCHEME for the June 2005 question paper

5090 BIOLOGY

5090/02 Paper 2 (Theory), maximum mark 80

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UNIVERSITY of CAMBRIDGE
International Examinations

JUNE 2005

GCE O Level

MARK SCHEME

MAXIMUM MARK: 80

SYLLABUS/COMPONENT: 5090/02

BIOLOGY
Paper 2 (Theory)

UNIVERSITY of CAMBRIDGE
International Examinations

Section A

1 (a) (i) plants/plausible named plant (R anything obviously not a water plant also R sea weed) ;

(ii) herbivore/consumer/2nd (trophic level) ; **[2]**
(R named)

(b) fertiliser/nitrates/salts/ions/CO_2/sewage/nutrients/manure ;
(A minerals, ignore warm water, mark first in list)
used for amino acids ;
for P/S AW (if CO_2 or HCO_3) ;
proteins ;
rapid AW growth (R growth unqualified) ;
greater reproductivity AW/population increases ; **[max: 4]**

(c) bacteria + decomposition/feed on dead plants or animals ;
bacterial population increases + plant population decreases ; **[2]**

(d) B - C line must rise to begin with ;
C → line must finish lower than it started and above arrow head ; **[2]**
(R line touching or crossing x axis)

[Total: 10]

2 (a) F, I, J (mark first 3) ;
G, H (mark first 2) ; **[2]**

(b) (i) (H) stigma + to catch pollen/ovary + to hold ovule (A female gamete)/seed ;
(A carpel/gynoecium/pistil)
(I) petal/corolla + to attract insect/landing platform ;
(J) stamen + to produce AW pollen/male gametes ; **[3]**
(A pollen sac/anther + filament or androecium)

(ii) large surface area/sticky/hairy/feathery ;
maximum AW + collection of pollen ; **[2]**

(c) line plus label L pointing to stigma on either F or H ; **[1]**
(R more than one line unless both correct,
A line to J if identified as stigma in **(b)**)

(d) ('It' = the pollen grain)
haploid AW/diploid AW/different no. of chromosomes ;
(correct ref. to) (ignore any stated numbers)
gamete/male contribution AW/ref. reproduction/meiosis
or/not involved in reproduction/part of adult plant/mitosis ; **[2]**

[Total: 10]

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3 (a) (i) (palisade) 1 cell drawn, correct shape and position and labelled ;

(ii) (spongy) 1 cell drawn, correct shape and position and labelled ;

(iii) 2 guard cells drawn in (lower) epidermis and labelled ;
(R label to stoma)

(iv) (cuticle) shown as thin acellular layer and labelled ;
(A cuticle on upper or lower epidermis, R if it crosses stoma)
(max: 2 marks if no labels) **[4]**

(b) xylem correctly indicated on Fig. 3.1 ;
xylem - the name (A even if incorrectly labelled) ; **[2]**

(c) greater concentration in the atmosphere ;
water molecules ;
loss of gradient AW ;
slow(er)/less evaporation ;
slow(er)/less diffusion ; **[max:3]**

(d) (Mark the first, but ignore waterproof)
make cuticle ;
protects photosynthesising or otherwise qualified cells ;
transparent/light entry ; **[max: 1]**

[Total: 10]

4 (a) (O) WBC/phagocyte AW (R lymphocyte or other BI) ;
(A polymorph/macrophage/cell membrane)
(P) capillary ; **[2]**

(b) ('It' = the blood vessel)
any two from :
walls + very thin/one cell thick ;
microscopic/narrow/small bore/allow RBCs only in single file ;
walls leaky/permeable AW/allow substances to pass through ;
(or named, A phagocytes AW)
slow blood flow ; **[max: 2]**

(c) (i) M - WBC/lymphocyte + antibodies ;
clumping AW (of bacteria) ; **[2]**

(ii) N - engulfing/ingesting/phagocytosis (R digesting) ; **[1]**

(d) action of platelets ;
fibrinogen to fibrin ;
clotting ;
trapping of RBCs ;
stopping bleeding ;
prevention of further bacterial entry ;
reproduction/growth of skin cells AW ; **[max. 4]**

[Total: 11]

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5 **(a)** **(i)** insulin (if wrong hormone, allow ecf* for **(ii)** - 1 mark only) ;

(ii) pancreas labelled (line must touch or enter gland) ;
[+*] (1 mark max. if more than one organ is named and labelled so long as one is the pancreas)
pancreas named (A even if incorrectly labelled) ;
(A islets of L)

(iii) any **two** (allowing ecf from either **(i)** or **(ii)** if applicable) from :
amylase, protease, lipase, alkali/salts, glucagons/CO_2
(A enzymes, pancreatic juice, but not with named enzyme or juice constituent) ;; **[5]**

(b) **(i)** bacterium/E.coli ; **[1]**

(ii) gene/sequence AW of bases ;
for insulin/hormone production ; **[2]**

(c) enzyme or named e.g. endonuclease(s), lingase(s)/catalyst ; **[1]**

[Total: 9]

[Maximum mark for Section A: 50]

Section B

6 (ref. to 'It' should be taken to refer to the first in each pair)

(a) no energy required in diffusion/diffusion is passive ;
energy required in active transport ;
ref. down a concentration gradient AW in diffusion ;
(R along)
ref. against concentration gradient AW in active transport ;
ref. respiration/ATP necessary in active transport/carriers ;
cell/living membrane in active transport ; **[max: 3]**

(b) (excretion) removal via lungs/bladder/skin ;
(A kidneys)
any **two** from: toxins, nitrogenous waste, CO_2 ;
(A urea/uric acid)
metabolic AW + waste AW ;
(egestion) removal from alimentary canal/rectum/anus/gut ;
undigested/e.g. cellulose/lignin/fibre/roughage/faeces ; **[max: 3]**

(c) (breathing) muscular ;
to move air/ventilate (R O_2, CO_2) ;
into and out of lungs/inhale and exhale ;
(respiration) in cells/metabolic AW ;
*oxidation/ref. oxygen + of glucose ;
*to release (or accepted alternatives) energy ; **[max: 4]**
(*available on an equation - balanced if symbols used)

[Total: 10]

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7 (A points on annotated genetic diagrams, and points, where relevant if they use the gene transfer idea in question **5**.)

(a)
sheep with better wool ;
used for reproduction/are crossed/selective breeding ;
gene passed to offspring ;
offspring vary ;
due to mutation ;
different gene combinations ;
'artificial selection' ;
process repeated ;
over many generations ;
use of back/test cross the find homozygotes AW ; **[max: 6]**

(b)
ref. mutation/ref. change in gene of chromosome ;
spontaneous/random/sudden/abrupt ;
mutagen/named mutagen (A radiation/chemicals) ;
meiosis (R spelling if a 't' appears) ;
random mating/cross fertilisation/recombination ;
ref. environment ;
one environmental factor identified ; **[max: 4]**

[Total: 10]

8 EITHER

(a)
protein ;
alters/speeds up rate of (chemical) reactions ;
in (living) cells/made by cells ;
catalyst/not used up/small amounts needed ; **[max: 3]**

(b) **(i)** or **(ii)** (pH or temperature)
ref. best/optimum/fastest ;
*rate slower both sides of the optimum ;
ref. active site AW ;
change in shape (of active site) ;
substrate no longer fits enzyme (or v.v.) ;
inactive outside range/range specific AW for enzyme ;

(i) (pH only)
*symmetrical curve or described ;

(ii) (temperature only)
destruction/denaturation only at high temperatures ;
(A 60 ^{o}C + if figure given)
(denaturing) permanent AW ;
heat increases rate of molecular movement or v.v/ref. Q_{10}/
increased frequency of collision AW (or v.v.) ; **[max: 7]**

(* = marks available on graph with axes labelled) **[Total: 10]**

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8 OR

(a) (i) (chlorophyll)
a green chemical/pigment/substance/molecule ;
contains magnesium ;
traps/harnesses/collects/harvests/absorbs/converts ;
light ; **[max: 3]**

(ii) (chloroplast)
structure/organelle ;
in plant/leaf + cells or named plant cell ;
contains chlorophyll ;
contains enzymes for photosynthesis ; **[max: 3]**
for part **(a)** **[max: 4]**

(b) (in **(i)** or **(ii)**) controlled by limiting factors or described

(i) (temperature)
*rate increases with increased temperature ;
faster molecular movement ;
photosynthesis is enzyme-controlled ;
ezymes work faster with increased temperature ;
may cause water loss slowing P/S ;
guard cells lose turgidity AW/stomata close (for **(b) (i)** max 4) ;

(ii) (light intensity)
*higher the light intensity the faster the rate of P/S ;
up to a maximum ;
more energy/light absorbed AW by chlorophyll/chloroplasts ; **[max: 6]**

(*marks available on graphs with axes labelled) **[Total: 10]**

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